



Integrating artificial intelligence into paediatric interventional radiology: a review of emerging applications and future directions

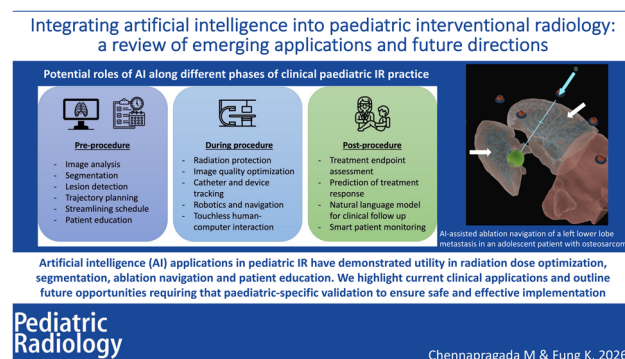
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Abstract

Artificial intelligence (AI) is heralded to revolutionise healthcare by improving efficiency, personalising care, and enhancing clinical outcomes. Although interventional radiology (IR) has been at the forefront of adopting AI technologies, their use within paediatric IR remains in early stages. Unlike adult practice, paediatric IR faces unique challenges, including scarcity of age- and pathology-specific datasets, wide anatomical and physiological variability across developmental stages, frequent need for sedation or anaesthesia, and heightened lifetime radiation risk. This review outlines the current and emerging applications of AI across the clinical paediatric IR pathway—such as pre-procedural planning, intra-procedural guidance, and post-procedural follow-up. We highlight areas where adult IR experience may inform paediatric adaption, while examining the technical, ethical, and regulatory considerations unique to children. By highlighting paediatric-specific needs and opportunities, we aim to delineate realistic pathways for safe, equitable, and clinically meaningful AI integration into paediatric IR.

Graphical abstract



Keywords Artificial intelligence · Children · Interventional radiology · Robotics

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Introduction

Artificial intelligence (AI) is increasingly transforming interventional radiology (IR) by enhancing procedural efficiency, image quality, workflow optimisation, and decision support. However, translation into paediatric interventional radiology represents distinct challenges, including limited age and pathology-specific datasets, anatomical and physiological differences, frequent sedation and anaesthesia requirements, and a higher lifetime radiation risk. In addition, the absolute number of paediatric patients requiring IR care is smaller

than adults. As a surrogate, the number of new cancer cases in the USA for children is nearly 15,000 compared to over 2 million of adults [1]. Consequently, compared with adult IR practice, the integration of AI-driven tools into paediatric IR remains at its early stages of development.

This review aims to provide a structured overview of current and emerging applications of AI in paediatric IR. The article is organised according to the key phases of the patient journey, as illustrated in Fig. 1. In addition, we discuss the potential role of AI in education and training, as well as the practical, ethical, and regulatory challenges associated with its implementation in paediatric IR. Fundamental concepts in AI, machine learning, and deep learning have been comprehensively reviewed elsewhere and are not reiterated here [2–5].

We highlight existing applications that are already in implementation in clinical practice with paediatric evidence and discuss future opportunities which require paediatric-specific validation for safe and effective implementation. To enhance clarity regarding clinical readiness, we distinguish throughout this review between AI applications that have undergone paediatric-specific validation and those that remain extrapolated from adult interventional radiology data. Where available, paediatric cohort data are highlighted; otherwise, applications are described as translational opportunities requiring dedicated paediatric validation. Commercially available or open-source AI solutions with current or potential relevance in paediatric IR are summarised in Table 1 [6–26].

Pre-procedural

Image interpretation, lesion detection, and planning

AI applications in imaging for interventional radiology have predominantly focused on image analysis tasks, particularly automated lesion detection, characterisation, and segmentation to support precise pre-procedural planning [27]. In

paediatric radiology, several studies have demonstrated the feasibility of AI-based detection systems across a broad spectrum of pathologies [28]. By leveraging deep learning algorithms to process complex imaging datasets, and extrapolating from adult data, AI can potentially identify subtle abnormalities that inform procedural planning for image-guided interventions such as percutaneous ablation [29] (Fig. 2).

Several AI tools originally developed for adult imaging—such as computer-aided detection systems for pulmonary nodules on chest radiographs of computed tomography—illustrate the potential for adaptation to paediatric oncology, where accurate identification of metastatic deposit is critical for pre-procedural patient selection and biopsy planning [30]. However, direct translation is not straightforward. Studies have demonstrated that an adult-trained AI system exhibits inferior performance for lung nodule detection in children, likely owing to smaller lesion size, greater respiratory motion artefact, and differences in thoracic anatomy and disease pattern [31, 32].

Beyond lesion detection, deep learning has also been applied to the automated identification and classification of tubes and lines on imaging, a task of particular relevance in critically ill paediatric patients [33]. The integration of imaging data with clinical, genetic, and biological information positions AI to perform comprehensive pre-procedural assessments—triaging patients for image-guided interventional procedures and even predicting therapeutic outcomes [34].

In the subspecialty of vascular anomalies, AI has shown promise in assisting in diagnosis, differentiation, classification, and segmentation of malformations based on combined clinical, dermatological, and imaging features [35]. Furthermore, machine learning has been applied to the management of arteriovenous malformations (AVMs), including prediction of angiographic architecture, risk stratification, and as a decision-support tool in stereotactic procedures [36]. However, many of these applications remain at the proof-of-concept stage and require prospective paediatric-specific validation.

Fig. 1 Schematic diagrams outlining the potential roles of artificial intelligence along different phases of clinical paediatric interventional radiology practice

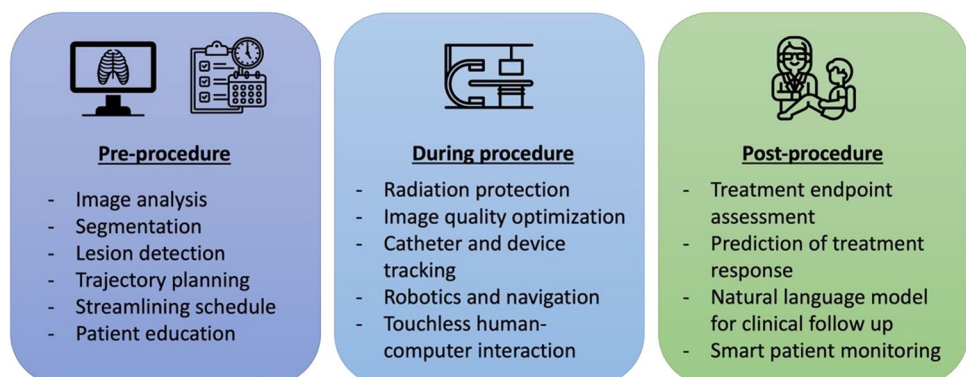


Table 1 Commercially available or open-source artificial intelligence solutions with current or potential application in paediatric interventional radiology

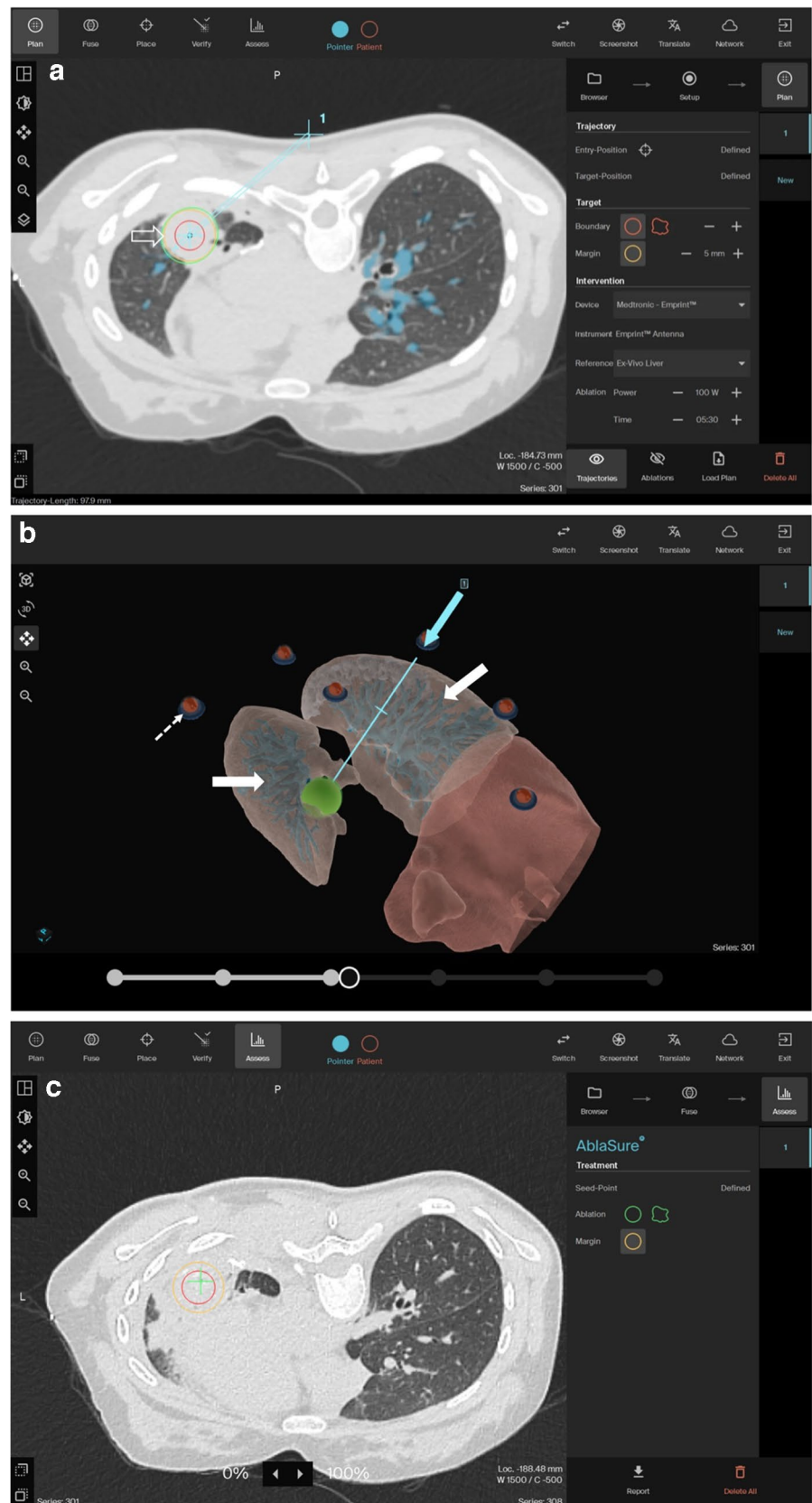
Current/potential application(s) in PIR	Example(s) of commercially available/open-source product(s)	Description of product	Regulatory clearance	Level of evidence in the literature	Paediatric-specific data
Virtual assistant for scheduling and administrative tasks	AI Assistant (Hyro, New York City, NY)	- Conversational AI chatbot which can help with patient scheduling, prescription refill and physician search - Quantitative and qualitative analyses of these conversations and patient cancellation can be performed via customised dashboards	Not available	Expert opinion [6]	No
Patient education	ChatGPT (OpenAI, San Francisco, CA)	- Provision of answers to questions covering various aspects of three interventional procedures (port implantation, percutaneous transluminal angioplasty, and transarterial chemoembolisation) including general information, procedure preparation and the procedure itself, and risks and complications as well as post-interventional aftercare - Transformation of CIRSE patient information leaflets available on CIRSE	Not available	Retrospective cohort [7]	No
Radiation dose optimisation and Image-denoising	Gemini 2.0 Flash (Google, Mountain View, CA)	- Regulation of acquisition parameters to automatically achieve optimal image contrast at patient radiation doses that conform to the As Low As Reasonably Achievable (ALARA) principle for radiation safety - Image post-processing to achieve reduce noise and improved image quality through automatic motion correction and real-time pixel shift	Not available	Conference proceedings [8]	Yes
	OPTIQ AI imaging chain (Siemens Healthineers, Erlangen, Germany)	Regulation of acquisition parameters to automatically achieve optimal image contrast at patient radiation doses that conform to the As Low As Reasonably Achievable (ALARA) principle for radiation safety	FDA approved	Expert opinion [9]	No
	ClarityIQ technology (Philips, Eindhoven, Netherlands)	Image post-processing to achieve reduce noise and improved image quality through automatic motion correction and real-time pixel shift	FDA approved and CE marked	Animal studies, retrospective cohorts [10, 11]	Yes

Table 1 (continued)

Current/potential application(s) in PIR	Example(s) of commercially available/open-source product(s)	Description of product	Regulatory clearance	Level of evidence in the literature	Paediatric-specific data
Augmented reality	OmnifyXR Interventional Suite (GE Healthcare, Chicago, IL/Mediview, Cleveland, OH)	- Holographic display of up to 4 video streams and a 3-dimensional anatomy model, via a headset - Voice commands can be utilised for hands-free interaction - Remote interface with mobile phones, tablets, or personal computers to allow education, proctor training, and teleconsultation	CE marked	Conference abstract, case report [12, 13]	No
Embolisation planning and vessel segmentation	Needle360 (Open source)	- Align percutaneous needle insertion site to specific angle for CT-guided biopsy of lung lesion - Automatic vessel segmentation from cone-beam CT - Simulation of superselective injection through virtual injection software - Vessel tracking and target planning	Not available	Conference abstract [8]	No
	EmboAssist AI (GE Healthcare, Chicago IL)	- Automatic vessel segmentation from cone-beam CT - Simulation of superselective injection through virtual injection software - Vessel tracking and target planning	FDA approved	Retrospective cohort [14, 15]	No
Ablation planning/confirmation support	BioTraceIO360 (Techsomed, Or Yehuda, Israel)	Planning, simulation, and confirmation of thermal ablation zone for liver, kidney, and lung lesions	FDA approved	Retrospective cohort [16]	No
CT-guided robotics/navigation	Ablasure® (Casination, Bern, Switzerland)	Automatic lesion detection, segmentation, and planning of needle path and needle targeting	FDA approved, CE marked	Case report, retrospective cohort [17–19]	Yes
	Cas-One® IR (Casination, Bern, Switzerland)	Automatic lesion detection, segmentation, and planning of needle path and needle targeting	FDA approved, CE marked	Case report, retrospective cohort, prospective multi-centre studies [17–20]	Yes
Endovascular robots	ANT-C (NDR Medical Technology, Singapore)	A single-use disposable and portable robotic system which enables manipulation of off-the-shelf catheters and guidewires	Not available	Animal and pre-clinical studies [21]	No
	LIBERTY® (Microbot Medical Ltd, Hingham, MA)	A single-use disposable and portable robotic system which enables manipulation of off-the-shelf catheters and guidewires	FDA approved	Animal study, prospective multi-centre single-arm study [22, 23]	No
Education and training	VIST® Virtual patient (Mentice, Gothenburg, Sweden)	Simulate real-life interventional scenarios with an immersive experience by integrating the virtual patient with angiographic machines	Not available	Retrospective and prospective cohorts [24, 25]	No

AI artificial intelligence, CE Conformité Européenne, CIRSE Cardiovascular and Interventional Radiology Society of Europe, CT computed tomography, FDA Food and Drug Administration, GE General Electric, GPT generative pre-trained transformer, IR interventional radiology

Fig. 2 Microwave ablation of a left lower lobe osteosarcoma metastasis in a 17-year-old girl. **a** During the planning phase, the artificial intelligence (AI) software detected the target left lower lobe lung metastasis (*open arrow*) in the pre-ablation axial contrast-enhanced computed tomography (CT) of the thorax in lung window, assisting the interventional radiologist to plan the biopsy trajectory. **b** The AI software was able to generate a real-time 3-dimensional volumetric rendering, indicating the needle path (*light blue*), left lower lobe metastasis (*green*), and segmented pulmonary vessels (*white arrows*). Note the surface trackers (*dashed arrow*) which were placed on the patient's skin and essential for the software to construct a 3-dimensional spatial map. **c** After ablation, the AI software was able to demonstrate the expected ablation margin on axial plain CT thorax in lung window (*orange circle*). Image courtesy of Dr. Simon Smith and Casination



Feasibility studies have also been performed on “home-grown” AI software and commercially available multi-modal large language models in planning of CT-guided lung biopsies in adult population [26, 37]. The 3-dimensional convolutional neural network (CNN) has achieved an overall AUC of 97.4% (95% CI, 96.3–98.2.3.2%) for lesion detection, with mean sensitivity of 93.5% and mean specificity of 93.2%. Among the software-proposed needle trajectories, 85.3% were feasible, with 82% matching actual paths and similar performance between supine and prone/oblique patient orientations [26]. These findings highlight the promising potential of AI-assisted segmentation, lesion detection, and trajectory planning for biopsy in children. In the future, integration of such planning tools with robotic guidance platform may pave the way towards increasingly automated, precision-driven biopsy procedures in paediatric IR, provided that safety and age-specific validations are rigorously addressed.

Scheduling and workflow optimisation

Patient scheduling plays a critical role in the operational efficiency, resource utilisation, cost-effectiveness, and overall patient experience within IR departments. AI and machine learning (ML) algorithms offer the potential to enhance scheduling by forecasting patient volumes, estimating visit durations and wait times, and dynamically adjusting appointment slots and staff allocation. These tools can also identify patterns in missed appointments and use demographic data to stratify risk across patient cohorts, thereby informing more targeted and efficient resource planning [38].

In paediatric IR, where procedural timing is tightly coupled with anaesthesia and nil-per-oral requirements, AI models can help optimise scheduling to reduce both fasting durations and room idling time. This is particularly valuable in large-volume centres, where minimising patient and caregiver distress is a key priority. In certain electronic health record systems, such as Epic (Verona, WI), predictive modelling tools are already used in adult surgical practice to identify underutilised operating room time and improve resource allocation efficiency [27, 39].

Beyond scheduling, AI can automate routine administrative tasks such as data entry, report generation, and communication with referring clinicians. When integrated into radiology information systems (RIS) and PACS platforms, AI enables continuous data analysis, automated prioritisation of urgent cases, and the generation of structured reports—ultimately reducing radiologist workload and enhancing workflow efficiency [40–42]. Commercially available AI-enabled virtual assistants, integrated with electronic medical record system such as Epic, are increasingly available to support scheduling coordination, patient verification, and prescription refill [6].

Despite the promise of AI in workflow optimisation, real-world adoption in paediatric IR practice remains inconsistent.

A recent metanarrative review highlighted considerable heterogeneity in how AI/ML models are developed and applied for scheduling [38]. Most existing evidence pertains to outpatient care, with limited data supporting their use in procedural or operating room settings. Nonetheless, the authors underscore the significant potential of AI-driven tools and call for further studies to assess not only their technical feasibility and effectiveness, but also their broader ethical and operational impact.

Anaesthetic/sedation risk prediction

Early exploratory studies in perioperative medicine have demonstrated the feasibility of ML algorithms in predicting recovery time after paediatric tonsillectomy and forecasting postoperative respiratory complications in adult surgical populations [43, 44]. Such predictive modelling has implications for procedural turnover efficiency, post-anaesthesia care unit bed allocation, and anaesthetic workforce planning. Translating similar tools to paediatric IR may improve peri-procedural flow while maintaining safety.

Patient education

AI can play an important role in helping patients and families better understand their medical conditions and procedures. Using natural language processing and large language models, AI can turn complex medical information into clear, easy-to-understand explanations. Recent studies have evaluated the use of commercially available large language models to provide in-depth patient education prior to IR procedures and procedural information for consent process [7, 45]. These models generally provided accurate information with improved readability without introducing potentially harmful misinformation.

In paediatric IR, however, educational needs are inherently dual-layered: information must be tailored to both the child—at an age-appropriate developmental level—and to the caregiver, who provides legal consent and ongoing support. AI-driven tools therefore hold particular promise in paediatrics, where they could generate parallel explanations by transforming more detailed procedural, risk and benefits information for parents or guardians, alongside simplified, reassuring narratives adapted for children.

For example, using Gemini Flash 2.0 (Google LLC, Mountain View, CA), patient information sheets issued by the Cardiovascular Society of Europe (CIRSE) were transformed to age-appropriate versions for children age 5–8 years; 9–12 years and adolescents. This approach resulted in a 71.0% improvement in mean Flesch Reading Ease scores (42.93 to 73.41, $P < 0.001$) and a 38.8% mean Flesch-Kincaid grade level (11.30 to 6.92, $P < 0.001$), indicating substantially enhanced readability (Fig. 3) [8]. By making procedural information more accessible, developmentally appropriate, and engaging, AI has the potential to

Fig. 3 Transformation of a (a) Cardiovascular and Interventional Radiological Society of Europe (CIRSE) patient information sheet on venous access port placement into (b) an age-adapted version for children aged 9–12 years using a large language model (Gemini 2.0 Flash, Google)

a

Original text from CIRSE: Venous access ports

What is a venous access port?

A venous access port is a central venous access device that allows doctors to easily access your veins to give treatments and to take blood. It is made of a non-irritant material and is designed to be inserted under your skin and remain in place for weeks or months. It is also known as a subcutaneous infusion port, and includes a catheter (a thin hollow tube), which is inserted through the skin and is then connected to a port in a pocket under the skin.

Venous access ports are commonly used in the care of cancer patients, but there are many other conditions that benefit from the use of venous access ports. Modern venous access ports are light and can be used during imaging procedures.

How does the procedure work?

The procedure for implanting a venous access catheter is performed on an out-patient basis, under fluoroscopic guidance. In most cases, the port is inserted into the patient's upper chest or arm. From there a thin catheter is placed in a vein.

Once the interventional radiologist has accessed the vein, they will use a guidewire to introduce a sheath and create a small pocket under the skin in the chest area. The catheter is then tunnelled to the vein and the port is connected to the catheter and placed in the pocket. Most physicians prefer to wait a week before starting to use the port. The wall of the port can be used for approximately 2000 punctures.

Why perform it?

The procedure is ideal for patients in need of long-term yet intermittent intravenous access. These patients typically receive chemotherapy or transfusions on a weekly or monthly basis and are unable to use a catheter inserted into a vein in the arm or hand.

Although the placement procedure is more complex and invasive than the more common technique of inserting a catheter into a vein in the hand or arm, central venous access allow patients to maintain daily activities, such as bathing, swimming and other forms of exercise. A venous access port has a lower risk of being dislodged than a catheter in the arm or hand. The port also requires fewer injections of heparin and fewer dressing changes. Because it is beneath the skin, it has an aesthetic advantage as well as a decreased risk of infection. Although venous access ports are expensive, the maintenance costs and risk of infection are low.

What are the risks?

Early complications are related to the technique itself, such as bruising, pneumothorax, nerve injury and an abnormal connection developing between an artery and a vein (called an arteriovenous fistula). Complications that may occur after the procedure include infection, a blockage or fracture in the catheter, blood clots and blockages in the vein.

Long-term complications include catheter fractures, which may lead to leakage under the skin, causing soft tissue death or wounds that do not heal.

If the venous access port is implanted under image guidance, the risk of procedure-related complications associated with surgical implantation is virtually eliminated.

help reduce anxiety, improve communication, and strengthen informed decision-making.

Intra-procedural

Radiation dose reduction and optimisation of image quality

Although many image-guided procedures in paediatric IR can be performed using ultrasound, fluoroscopy and CT remain important contributors to cumulative radiation

exposure. Dose reduction is particularly critical in children because of their increased radiosensitivity, longer life expectancy, and greater lifetime risk of radiation-induced malignancy [46]. These considerations underpin the As Low As Reasonably Achievable (ALARA) principle, which mandates optimisation of radiation dose without compromising diagnostic or procedural efficacy.

To support this goal, vendors have integrated tools such as radiation dose structured reports and software-based dose modulation into angiography and fluoroscopy systems to facilitate quantitative dose monitoring and optimisation [47]. In parallel, AI-driven imaging strategies have

Fig. 3 (continued)

b LLM Transformed Text (for 9 – 12 years old)

What is a venous access port?

Imagine having a little helper button under your skin that makes it super easy for doctors to give you medicine or take a blood sample! That's kind of what a venous access port is. It's a small, round device made of special material that won't bother your body. It sits just under your skin and is connected to a thin, soft tube (like a tiny straw) called a catheter that goes into one of your big veins.

Think of it like a secret door to your veins! It lets the doctors get to your blood without having to poke you with a needle every time. It can stay in place for weeks or even months, so you don't have to keep getting stuck. They're also sometimes called "subcutaneous infusion ports".

How Does it Get Put In?

Getting the port put in is like a little construction project under your skin. You'll come to the hospital, and the doctor will make sure you are comfortable. Most of the time, the port goes in your upper chest or arm.

To help them see inside your body as they work, the doctor will use a special X-ray machine that shows pictures on a screen, like watching a live video of what's happening inside. They will make a small pocket under your skin and gently thread the catheter into your vein. Then, they connect the port to the catheter and tuck the port into the pocket.

After the port is placed, the doctor might want to wait about a week before using it, so it has time to settle in. The top of the port is like a special, strong spot that can be poked with a needle many, many times!

Why Do I Need One?

This special port is great if you need medicine or blood tests often, like if you have to get treatments every week or month and it's hard to use veins in your arm or hand.

Even though it's a little more work to put in the port than just using a needle in your arm, it lets you do more of your normal activities, like taking baths, swimming, and playing sports! It's also less likely to get pulled out than a regular IV, and it needs fewer bandages. Plus, because it's under your skin, it looks nicer and is less likely to get infected.

Are There Any Risks?

Like with any procedure, there are a few possible risks, but the doctors and nurses will do everything they can to keep you safe. Some things that could happen early on are bruising or, very rarely, a problem with a nerve or blood vessel. Later on, there's a small chance of getting an infection, a blood clot, or the catheter getting blocked or broken.

If there are any problems with the catheter, it can sometimes leak under the skin. But the doctors are very careful when they put the port in, using the X-ray machine to guide them, which makes these problems much less likely.

emerged as additional tools for dose reduction, aligning with the objectives of the Image Wisely and Image Gently campaigns, which advocate minimising unnecessary radiation exposure, tracking dose indices, and optimising modality selection in children. These include automated protocol selection based on patient size and clinical indication, low-dose preset configurations, and deep learning-based image reconstruction techniques for paediatric CT [48]. In paediatric imaging, MacDougall et al. demonstrated a 31% reduction in image noise by training a neural network to generate iteratively reconstructed CT images

from filtered back projection data of low-dose abdominal CT scans [49]. Notably, radiologists showed a preference for the AI-reconstructed images over conventional iterative reconstructions.

In fluoroscopy, AI-enabled ultrafast collimation systems dynamically confine radiation exposure to the region of interest (ROI) while shielding adjacent tissues using a secondary collimator. High-quality real-time imaging is maintained within the ROI, and previously acquired full-field images are superimposed to preserve anatomical orientation. This selective exposure is typically imperceptible

to the operator. Such auto ROI processor is trained using a CNN-based algorithm to recognise key components within the ROI. Clinical studies have demonstrated radiation dose reduction of up to 60% for both patients and operators compared with conventional fluoroscopy platforms [50]. Notably, current clinical validation has been performed in adult interventional cardiology and endoscopy, and paediatric-specific outcome data are not yet available. Potential applications in PIR include endovascular intervention in portal venous system and neurological diseases where radiation exposure can be significant [51, 52].

Digital subtraction angiography (DSA) enables superior visualisation of vascular anatomy; however, its radiation dose is relatively high. In a study using anthropomorphic phantoms, the authors showed that in a typical cerebral angiography procedure, the radiation dose from DSA is nearly three times that from fluoroscopy [53]. Commercially available image-processing technologies have demonstrated meaningful dose reductions in paediatric interventions. In one paediatric-specific study evaluating 21 types of peripheral and abdominal interventional procedures, implementation of a novel real-time post-processing algorithm reduced the third-quartile procedure dose–area product (DAP) by 30% (standard technology, 4.89 Gy·cm²; novel technology, 1.45 Gy·cm²) and procedure air kerma by 36% (standard technology, 33.26 mGy; novel technology, 12.0 mGy) [11]. Many AI-based radiation dose optimisation tools are already incorporated into commercially available angiography platforms and arguably represent the most practice-ready applications for paediatric interventional radiology (Table 1). Importantly, the interventional radiologists should ensure paediatric-specific low-dose protocols and size-adjusted settings are actively selected, as default system parameters may not automatically apply dose-saving features tailored for children.

Recent advances in AI have enabled deep learning subtraction angiography—generation of synthetic high-quality, DSA-like images using native angiogram as input [54]. This eliminates the need for a preliminary mask acquisition, thereby reducing radiation dose and avoiding motion-related artefacts caused by inter-scan misalignment. Its performance in paediatric patients, particularly in neonates and young children, remains unproven. It is reasonable to postulate that smaller vessel calibre may potentially lead to higher rate of misalignment artefact [55].

Real-time analysis of imaging acquired during angiography has been used to estimate and analyse fractional flow, a useful indicator for ischaemia in coronary artery disease [56]. A similar application may be useful in other regions for flow assessment of paediatric vascular diseases, including renal artery stenosis or Moyamoya disease. It may aid in the selection of appropriate devices such as stents or embolisation coils [57].

Automatic vessel analysis is already widely integrated into most contemporary angiographic systems. More recently, novel AI algorithms have enabled automatic vessel segmentation from cone-beam CT acquisitions, facilitating 3-dimensional vascular mapping during procedures [15]. Complementary virtual injection software allows simulation of selective catheterisation from a chosen vascular pedicle, enhancing procedural planning. Adult studies have demonstrated that these tools can improve pre-procedural strategy, reduce contrast utilisation, and lower radiation exposure [14]. While no dedicated paediatric studies have been published to date, the authors have used these functions in children to map angioarchitecture for embolisation of tumour or high flow vascular malformations. The extension of deep learning to real-time analysis of vessel segmentation on intravascular ultrasound (IVUS) may enable more precise device sizing and placement [58].

Fusion imaging has become an important tool in interventional radiology, allowing for more precise targeting and navigation during complex procedures. By overlaying pre-acquired CT or MRI datasets onto real-time ultrasound or fluoroscopic imaging, fusion techniques provide enhanced anatomical context and procedural accuracy. AI can further improve the accuracy of image registration, reducing spatial misalignment and enhancing the fidelity of fused images. In contrast to conventional co-registration methods, which rely on alignment using anatomical landmarks, AI applications can perform deformable, nonlinear registration to account for motion artefacts, vessel pulsation, and anatomical distortion. While no paediatric-specific validation is currently available, a recent consensus statement from the World Federation of US in Medicine and Biology confirms that AI and ML will play an important role in optimising image co-registration in the coming years [59, 60].

Catheter and device tracking

CNNs have shown significant promise in enhancing device localisation during image-guided procedures. For ultrasound-guided interventions, Mwikirize et al. employed a faster region-based CNN (R-CNN) to accurately visualise needles of varying sizes and insertion depths in porcine and bovine models, achieving an impressive precision of 99.6% [61]. In fluoroscopic imaging, AI can assist with the reliable visualisation of interventional devices—an essential function during complex vascular procedures. Ghosh et al. demonstrated the utility of a topography-aware geometric deep learning model in detecting catheter tips and supporting catheter segmentation during cerebral angiography [62]. To date, validation studies have been only conducted in adult populations but not in paediatric populations.

In clinical practice, tools such as Neurovascular Assist (iMed Technologies, Tokyo, Japan) have introduced

real-time AI-assisted tracking of devices including guidewires, guiding catheters, embolic materials, and protection devices used in neurovascular procedures. The software issues alerts when devices shift position, enhancing intra-procedural awareness. Two small clinical series demonstrated both the safety and high notification accuracy of this system—one focusing on device tracking, and the other evaluating its performance with liquid embolic agents [63, 64]. Such models derived from adult population may hold translational potential to the paediatric population; however, validation in a younger cohort is essential as the smaller vessel calibres in children may challenge segmentation performance [65].

Robotics and navigation

Robotic assistance is well established in the surgical field and is now emerging as a promising adjunct in percutaneous and endovascular interventional procedures. In a recent study, Bonnet et al. (2024) reported the safe and effective use of robotic systems for CT-guided thermal ablation of abdominal tumours in 41 patients [66]. Robotics in interventional radiology offers several advantages, including enhanced precision, improved technical capabilities, and reduced radiation exposure. When combined with artificial intelligence, robotic platforms hold the potential to evolve into fully autonomous systems capable of performing complex procedures with minimal operator input.

In the endovascular space, robotic-assisted interventions are gaining traction, particularly in cardiovascular and neurovascular procedures. The CorPath GRX (Siemens Healthineers, Erlangen, Germany) robotic platform has been utilised in the endovascular embolisation of cerebral aneurysms [67], while the ROSA system (Zimmer Biomet, Warsaw, IN), originally developed for intracranial targeting, was successfully used for transtemporal embolisation of a carotid-cavernous fistula [68]. Other commercially available systems—such as the Magellan Robotic System (Johnson & Johnson, New Brunswick, NJ), Robocath R-One (Johnson & Johnson, New Brunswick, NJ), and Sensei X (Johnson & Johnson, New Brunswick, NJ)—have demonstrated technical precision that is comparable to, or in some instances exceeds, that of manual catheter manipulation [69]. Emerging systems such as LIBERTY (Microbot Medical, Hingham, MA) are designed as single-use, portable platforms, facilitating ease of transportation, improved ergonomics, and reduced cross-contamination risks. Notably, LIBERTY is compatible with standard off-the-shelf instruments, eliminating the need for proprietary catheters or guidewires and potentially lowering barriers to clinical adoption [23].

Beyond vascular applications, several non-vascular robotic systems—including AcuBot, Innomotion, and Epione—have shown high precision in CT- and

fluoroscopy-guided biopsies and ablations [58]. Notably, two case reports have described the successful use of the CAS-One (Casination, Bern, Switzerland) navigation system to guide CT-guided ablation procedures in paediatric patients [18, 19]. Such a system can eliminate the need for continuous needle position correction, thereby reducing radiation dose and improving the precision of needle placement. However, widespread adoption of robotic systems in both paediatric and adult interventional radiology remains constrained by several challenges, including limited haptic feedback, restricted real-time manual override capability, and a paucity of large-scale prospective clinical data.

A particularly promising area for paediatric interventional radiology is the development of MRI-compatible robotic systems. These platforms offer the ability to perform interventions with real-time functional imaging, providing valuable physiological information such as blood flow, perfusion, and oxygenation—factors that are especially important in delicate paediatric interventions [58]. At present, to our knowledge, no commercially available MRI-compatible robotic endovascular system has reached the market. Most existing platforms remain in the pre-clinical or early translational development phase, with ongoing efforts focused on overcoming challenges related to magnetic compatibility, device miniaturisation, actuation mechanisms, and workflow integration within the MRI environment.

Touchless software interaction and augmented/virtual reality

In the IR suite, physicians often encounter challenges when interacting with computer systems while maintaining a sterile environment. Traditional input methods, such as keyboards and touchscreens, are impractical during procedures and can introduce inefficiencies or risks. Touchless human-computer interaction (HCI) technologies offer a promising solution by enabling clinicians to interact with medical software through voice and gesture-based commands (Fig. 4, Supplementary material 1), thereby preserving sterility and enhancing procedural workflow [70]. In one of the authors' institution, such touchless system has been used to manipulate and cross reference imaging studies to maintain sterility during the procedure.

ML plays a crucial role in enabling these touchless interactions. ML algorithms can be trained to recognise and classify spoken commands and hand gestures, improving the accuracy and responsiveness of cameras, microphones, and other sensor-based devices [71, 72]. By facilitating smoother, more intuitive communication between the operator and the system, these AI-enhanced touchless interfaces can reduce cognitive burden, minimise procedural interruptions, and contribute to safer and more efficient interventional care.

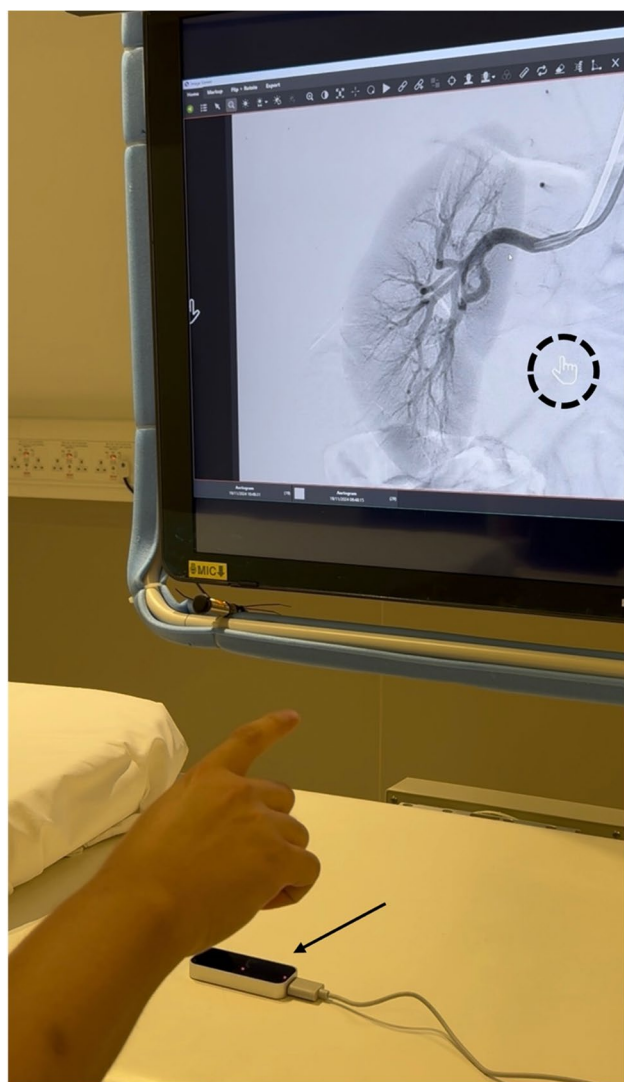


Fig. 4 Touchless software interaction. Photograph showing the use of an infra-red sensor (*black arrow*) to detect the operator's hand gesture, enabling interaction with the medical image viewer while maintaining sterility. The display shows a digital subtraction angiography of right kidney of a 10-year-old boy, with a virtual cursor (*dashed black circle*) controlled through gesture-based input

Beyond touchless control, augmented reality (AR) platforms are emerging within the next-generation interventional suites. The OmnifyXR Interventional Suite (GE Healthcare, Chicago, IL/Mediview, Cleveland, OH) incorporates holographic heads-up display technology that streams live imaging with 3-dimensional anatomical visualisation. Although currently available only in limited centres in the USA, such platforms hold potential for enhanced spatial orientation, improved procedural planning, and even remote collaboration. In regions where paediatric IR expertise is limited, AR-enabled systems may facilitate real-time consultation and knowledge sharing, potentially expanding access to specialised care.

Post procedural

Outcome prediction

AI-powered decision-support systems leverage advanced ML algorithms to predict disease progression and post-intervention outcomes. Unlike traditional predictive models that rely on linear relationships and limited datasets, ML techniques can identify nonlinear associations among a wide range of clinical, imaging, and radiomic variables to generate personalised, patient-specific predictions [73]. Most current evidence derives from adult cohorts. In interventional oncology, such models have been applied to evaluate ablation completeness through automated segmentation of the ablation margin [74] (Fig. 2c) and to predict treatment response following transarterial chemoembolisation (TACE). Similar techniques have also shown promise in mortality prediction after stroke and in forecasting outcomes following treatment of arteriovenous malformations [75].

A particularly compelling future direction is the application of AI to predict vascular and somatic growth trajectories in children, thereby informing optimal sizing of stents and implantable devices (e.g. central venous catheters, ureteral stents, gastrojejunostomy tubes). Accurate growth modelling could reduce device-related complications, minimise the need for early reintervention, and limit material wastage. Early adult-focused work has already explored AI-based estimation of biliary stent length, providing a conceptual framework for adaptation to paediatric interventional practice [76].

The ability of AI to process and integrate diverse patient datasets—including electronic health records, imaging studies, and genomic information—enables the development of more precise diagnostic tools and personalised therapeutic strategies. For example, radiogenomics has shown utility in prognostication among oncology patients with hepatocellular carcinoma (HCC), renal cell carcinoma (RCC), and colorectal cancer (CRC) [77]. In the vascular anomalies domain, radiomics-based ML models have been used to distinguish fibroadipose vascular anomalies from venous malformations with high accuracy [78]. Such models could be adapted for paediatric interventional radiology (PIR) to support diagnosis and treatment planning for complex vascular anomalies.

Another valuable application of AI lies in data augmentation and artificial dataset generation [79]. This is particularly relevant in paediatric IR, where data scarcity—especially for rare diseases and low-incidence conditions—limits model development. AI-generated synthetic datasets can help train more robust and generalisable models, mitigating the limitations of small sample sizes and enhancing predictive performance for complex interventions in young patients.

Education and training

AI-powered tools can be used to enhance training methods, particularly for radiology residents, by providing a simulated environment. Real-time feedback with precise and objective evaluation aids in accelerating the acquisition of skills. The utility of AI in surgical education has been well explored. A scoping review by Kirubarajan et al. identified 49 studies relevant to artificial intelligence in surgical education and found a high degree of accuracy in objectively categorising surgical skill sets [80]. Most frequent AI interventions used artificial neural networks and support vector machine, among other supervised learning models. It is important to note that despite the promising outcomes that many of these studies have shown, most of them have been proof-of-concept studies, and few have been incorporated into regular surgical training curricula. This may reflect, to a certain degree, a lack of trust in computer-based interventions. Nonetheless, these results hold promise for AI-driven training and evaluation models, which can play a significant role in procedural training in fields such as interventional radiology [73, 81].

Virtual reality simulations provide a fully immersive environment created for procedural training, allowing for optimal hands-on training in a safe and effective environment. Building simulation databases also provides for training on a broad case mix and allows for the development of diverse skill sets. While current AI-driven solution is scarce, a generative adversarial network has potential to generate virtual scenarios in a risk-free environment. This is of particular importance in subspecialties such as paediatric IR, where the real-life volume of complex cases may be limited.

Virtual reality (VR) simulation offers an immersive and safe environment for procedural training in PIR, where real-life case volumes—particularly for rare or complex conditions—may be limited [29, 79]. Although AI-driven simulation tools remain relatively underdeveloped, generative adversarial networks (GANs) have shown promise in creating realistic synthetic procedural images, including ultrasound and angiographic simulations with authentic anatomical texture and contrast dynamics [82]. By generating controlled variations of scenarios such as training modules for radiation protection, selection of arterial access in children, or procedural complications, GAN-enhanced platforms could broaden trainee exposure in a risk-free and standardised manner. Simulating scenarios in neonatal population may be particularly helpful for trainees to learn to limit contrast use and navigation of catheters in small-calibre vasculature. However, while synthetic data may help address case scarcity, rigorous validation is essential in paediatric IR to ensure clinical fidelity and avoid artefacts.

Challenges

The biggest challenge for the implementation of machine learning in IR is the insufficient data for building robust AI models. High-quality, sufficiently large datasets are complex to obtain for real-time interventional imaging. Heterogeneity across the available data, such as anatomical variability and diverse pathology, especially in paediatric IR, along with the inconsistencies in imaging protocols and non-standardised techniques, adds to the challenges. In children, these challenges are amplified by smaller anatomical structures and wide physiological variation across developmental stages, which may substantially affect algorithm performance.

As an increasing number of AI/ML models are being explored in the medical domain, the issue of potential bias becomes vital to avoid unfair and detrimental application of AI. This is again particularly relevant in paediatric medicine. Only 17% of FDA-approved medical devices are approved for paediatric use [83]. Hua et al. found that children are systematically underrepresented in public medical imaging data, making up less than 1% of datasets [84]. This precludes reliable AI models from being developed for paediatric applications. Caution must be applied when attempting to adapt adult-trained AI system for paediatric use. A recent scoping review showed that most adult-trained AI tools showed performance reduction when directly applied in children [85]. Careful model validation must be performed before clinical implementation.

Furthermore, ethical concerns arise regarding patient data privacy when large amounts of sensitive personal health information are made available for the development of AI and ML models. Additionally, the implications of applying AI in vulnerable populations, such as children, underscore the need for robust governance frameworks and data-sharing processes to be established [86, 87].

Finally, adoption barriers extend beyond technical limitations. Retention for adopting emerging technologies can be a significant obstacle due to inherent scepticism of clinicians who may find AI-driven systems less reliable and untrustworthy [88]. Similarly, parents may be hesitant to provide consent for children to undergo AI-driven interventions due to risk aversion to novel technologies and lack of robust paediatric-specific evidence.

Conclusion

Rapid advancements and continuous improvements in technology have led to the emergence of AI as a transformative tool in procedural medicine, including interventional radiology. Real-time image enhancement, AI-driven navigational

tools, and clinical decision-support systems can significantly improve the precision, safety, and efficacy of interventional procedures, while optimising patient outcomes. Fields such as paediatric interventional radiology can immensely benefit from the AI's ability to optimise equipment, reduce radiation exposure, customise procedures, and provide augmented datasets for training and research.

Despite this promise, the full potential of AI remains underutilised in real-world clinical practice. Challenges, including data scarcity, algorithmic bias, and ethical concerns related to data privacy and governance, continue to hinder widespread adoption—issues that are especially pertinent in paediatric populations. The establishment of multi-centre paediatric IR registries would facilitate robust data aggregation across institutions, while federated learning approaches may enable collaborative model training without compromising patient privacy [89]. Close collaboration among paediatric radiologists, interventionalists, data scientists, engineers, and regulatory bodies will be essential to ensure safe validation, equitable deployment, and appropriate oversight. Through such collective efforts, AI can be responsibly integrated into paediatric IR, translating technological innovation into clinical benefit for children and their families.

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- All procedures performed in this report were in accordance with the ethical standards of the institutional review board and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.
- This article does not contain any studies with human participants or animals performed by any of the authors. Formal Institutional Board Approval is waived for this type of study.
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